Can We Predict Which Startups Win?

54,294 startups · 3 outcomes · 5 notebooks

The Journey

```
flowchart LR
    A["📊 EDA 01"] --> B["🧪 Research Questions 02"]
    B --> C["⚙️ Preprocessing 03"]
    C --> D["⚖️ Imbalance Handling 04"]
    D --> E["🤖 Feature Selection & Modeling 05"]
```

Each notebook builds directly on the one before — decisions made in `01_EDA` shaped the pipeline in `03_Preprocessing`, and the resampling choices in `04` carried into every experiment in `05`.

Phase 1 — EDA

`01_EDA.ipynb`

The raw data had widespread nulls and no ready-to-use features. Before writing a single line of preprocessing code, we mapped every missing value pattern and decided how to handle it — drop, impute, or encode as "Unknown." At the same time, we identified which raw columns could be transformed into informative features: binary round flags, log-scaled funding amounts, geographic clusters, and target-encoded categories.

```
pie title Startup Outcomes
    "Operating" : 86.5
    "Acquired" : 8.1
    "Closed" : 5.4
```

Funding amounts — before and after log transform:

```
LINEAR SCALE
seed  ████████████████████████████████████████░░░░    99% near zero, spikes at tail

LOG SCALE
seed       ░░░░████████████████████░░░░               near-normal distribution
```

Null & Feature Engineering Decisions

#	Problem Found	Decision Made
1	**15K+ rows missing founding date**	Median imputation per column (year / month / quarter)
2	**Geography dominated by USA**	Silicon Valley, NY, Boston, Seattle, LA → 7 named clusters
3	**753 markets, 115 countries**	One-hot encoding impractical → target encoding (mean acquisition rate)

4	`funding_total_usd` **vs** `rounds_sum` **mismatch**	Cross-referenced to separate true zeros from missing data
5	**Round amount columns mostly zeros**	Converted to binary flags: `has_round_A / B / C`
6	**Rounds D–H each below 3% participation**	Too sparse individually → collapsed to `round_D_plus`

Phase 2 — Research Questions

`02_Research_Questions.ipynb`

We started with open, curious questions about what makes startups succeed — the kind you'd argue about over coffee. We then realized we could actually answer them with the data alone, no model needed.

```
flowchart TD
    RQ1["More seed\nfunding?"] --> A1["✅ 6% → 19%\nacquisition rate"]
    RQ2["Industry?"] --> A2["✅ Semiconductors\n2–3× average"]
    RQ3["Location?"] --> A3["✅ SV ~15%\nnon-USA ~3%"]
    RQ4["Angel / Grant?"] --> A4["⚠️ Angel +7pp\ngrants weak"]
    RQ5["Speed to funding?"] --> A5["✅ Under 3 months\n→ 11% acq rate"]
    RQ6["Founding era?"] --> A6["⚠️ Crisis cohort\nlower — but biased"]
```

Bias	Problem
Survivorship	Pre-2000 startups look best — failures were deleted from the dataset
Immaturity	2012+ startups not old enough to be acquired yet
Causality	Correlation only — we can't say seed funding *causes* success

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Phase 3 — Preprocessing

`03_Preprocessing.ipynb`

Every decision from the EDA was translated into a step in a reproducible sklearn Pipeline — so the exact same transformations apply to train, validation, and test without any leakage.

```
flowchart LR
    IN["39 cols nulls everywhere"] --> P1["Fix dtypes"] --> P2["Parse dates"] --> P3["Fill unknowns"]
    P3 --> P4["Geo clusters"] --> P5["log1p funding"] --> P6["Round flags"]
    P6 --> P7["Angel flag"] --> P8["Never-funded flag"] --> P9["Days to funding"]
    P9 --> P10["Target encode"] --> P11["Median impute dates"] --> P12["Drop raw cols"]
    P12 --> OUT["46 features 0 nulls"]
```

Split	Rows	Share

Train	27,860	70%
Validation	5,971	15%
Test	5,971	15%

> *Pipeline fitted on training data only — applied identically to validation and test.*

Phase 4 — Imbalance Handling

`04_Imbalanced_Data_Handelling.ipynb`

The preprocessed training data had a 15:1 majority/minority ratio — a model with no intervention would simply learn to predict "operating" for everything. We evaluated five strategies to correct this before any model training began.

```
BEFORE RESAMPLING
Operating  ████████████████████████████████████████████  86.5%
Acquired   ████░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░    8.1%
Closed     ██░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░░    5.4%

AFTER SMOTETomek
Operating  ██████████████░░░░░░░░░░░░░░░░░░░░░░░░░░░░   32.8%
Acquired   ██████████████░░░░░░░░░░░░░░░░░░░░░░░░░░░░   33.6%
Closed     ███████████████░░░░░░░░░░░░░░░░░░░░░░░░░░░   33.6%
```

Strategy	Macro F1	Accuracy	Train Rows
Baseline (none)	0.387	86.2%	27,860
ROS	0.452	64.9%	~72,000
RUS	0.422	59.1%	~4,500
SMOTE	0.466	83.9%	62,364
SMOTETomek	**0.479**	**83.8%**	**62,364**
Class Weights	0.439	84.9%	27,860

All 5 variants carried forward into modeling.

Phase 5 — Feature Selection & Modeling

`05_Feature_Selection_n_Modeling.ipynb`

We ran a democratic feature vote across 8 methods to select the 24 most informative features, then combined them with the full experiment grid — every model against every resampling strategy — to find the best combination. The test set was used exactly once at the end.

```
flowchart LR
    F["32 features"] --> V1["ANOVA"]
    F --> V2["Chi²"]
```

```
    F --> V3["Random Forest"]
    F --> V4["Gradient Boosting"]
    F --> V5["Lasso"]
    F --> V6["Ridge"]
    F --> V7["SVM"]
    F --> V8["Forward Select"]
    V1 & V2 & V3 & V4 & V5 & V6 & V7 & V8 --> VOTE["Vote ≥ 5/8 + corr pruning"]
    VOTE --> OUT["24 features selected"]
```

The full experiment grid:

	All Features (32)	Selected Features (24)
Logistic Regression L2	× 5 resampling	× 5 resampling
Logistic Regression L1	× 5 resampling	× 5 resampling
Random Forest	× 5 resampling	× 5 resampling
Gradient Boosting	× 5 resampling	× 5 resampling
XGBoost	× 5 resampling	× 5 resampling

5 models × 2 feature sets × 5 resampling strategies = 50 experiments

Primary metric: **Macro F1** (all three classes weighted equally). Test set touched exactly once.

Key Lessons

- **Feature engineering beats model choice** — the 39 → 46 feature transformation mattered far more than which algorithm was used
- **Imbalance follows you everywhere** — resampling in `04` was only the beginning; every modeling decision downstream depended on it
- **Correlation is not causation** — we predict patterns, not mechanisms; that's an honest scope

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